Adjusting the Probability of Direction for Multiple Testing

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Abstract

We examine the use of the probability of direction (pd) as a decision rule in psychological research, where results are often interpreted using conventional thresholds (e.g., $pd > 0.975$). When such thresholds are applied repeatedly across multiple parameters, a multiplicity problem arises, as the probability of obtaining at least one misleading conclusion increases with the number of tests. We characterize the operating properties of pd-based decisions under repeated use, exploiting its formal correspondence with the frequentist p-value in common settings. Through simulation, we show that extreme pd values can arise by chance at non-negligible rates, leading to inflated family-wise error rates. To address this issue, we recast pd in terms of posterior odds and derive a prior-odds adjustment that accounts for multiplicity by down-weighting the prior probability assigned to each individual hypothesis. This approach separates within-model priors from prior assumptions over the hypothesis space, allowing researchers to control error rates while maintaining flexibility in prior specification. The method is extended to account for correlations among tested hypotheses and is benchmarked against standard frequentist corrections. An accompanying R package implements the proposed procedure, facilitating its use in applied research.

Keywords: probability of direction, multiple testing, prior-odds

Adjusting the Probability of Direction for Multiple Testing

Over the past two decades, the use of Bayesian methods in psychological research has grown considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; Van De Schoot et al., 2017). Unfortunately, this expansion has been accompanied by practices reminiscent of the "null ritual" in frequentist statistics, where widespread use was associated with automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). For instance, researchers sometimes treat Bayes Factor thresholds (e.g., BF > 3) as dichotomous decision rules, or claim that an effect "exists" because the posterior probability of a directional effect exceeds a conventional benchmark such as 95% (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). While such summaries are convenient, their use as default cutoffs encourages all-or-none interpretations. When these binary judgments are applied repeatedly across many parameters, the issue of multiplicity arises.

In psychological research, the problem of multiplicity is considered selectively rather than systematically. Researchers typically adjust for multiple comparisons in well-recognized cases, such as post hoc comparisons, but rarely do so when evaluating several effects simultaneously within the same model, as is routine in ANOVA or multiple regression (e.g., main effects and interactions, or multiple predictors) (Bender & Lange, 2001). Yet each of these corresponds to a distinct hypothesis test. When each test is evaluated using a fixed evidential threshold, the probability of obtaining at least one misleading result increases with the number of tests (Cramer et al., 2016). This implies that researchers effectively adopt a higher overall error rate than intended, contributing to an inflated literature and undermining replicability (Zeevi et al., 2025).

The frequentist framework formalizes this as a Type I error problem: if each test is conducted at level α, the probability of at least one false positive grows with the number of tests. Standard approaches therefore control either the family-wise error rate (FWER) or the false discovery rate (FDR) (Benjamini & Hochberg, 1995; Westfall & Young, 1993). These procedures address whether a spurious result enters the literature, but leave unresolved a subtler consequence of the same selection process: how large reported effects appear. When researchers examine many

parameters but highlight only the most extreme estimates, selection systematically inflates reported magnitudes, a phenomenon known as Type M error (Altoè et al., 2020; Gelman & Carlin, 2014; Zwet & Cator, 2021).

Bayesian methods address Type M error by shrinking extreme estimates toward more plausible values, either through hierarchical models with partial pooling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011) or informative priors that reduce overestimation in low signal-to-noise settings (Dawid, 1994; Zwet & Gelman, 2022). Crucially, however, this does not resolve the multiplicity problem entirely. When the same decision threshold is applied to many parameters, for example, declaring a positive effect whenever the posterior probability exceeds 95%, the probability of at least one spurious conclusion can remain substantial. Bayesian solutions to this problem rely on either simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020) or priors over the hypothesis space that scale with the number of tests (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997). Despite these available solutions, hierarchical modeling is still often treated as sufficient for multiplicity control (Chen et al., 2020; Erčulj & Štrumbelj, 2015; Hogeveen et al., 2022; Limbachia et al., 2021; Mathar et al., 2022).

One might object that this concern does not apply under a subjectivist interpretation of Bayesian inference, which treats posterior probabilities as rational degrees of personal belief and therefore not subject to frequentist error criteria (Edwards et al., 1963; Lindley, 1972; Rozeboom, 1960). This argument, besides calling for fully coherent joint inference rather than marginal summaries, sits uneasily with standard psychological practice, where researchers typically apply default weakly informative priors rather than priors encoding substantive domain knowledge (Smid & Winter, 2020; Van De Schoot et al., 2017). With such priors and sufficient data, Bayesian and frequentist procedures often converge (Albers et al., 2018; Bernardo & Smith, 1994; Gelman et al., 2013).

Rather than adjudicating whether Bayesian analyses should be held to frequentist standards, we therefore adopt a falsificationist approach (Box, 1980; Gelman & Hennig, 2017;

Gelman & Shalizi, 2013; Mayo, 2018) and ask a more tractable question: what are the operating characteristics of a widely used Bayesian decision rule, and how can its multiplicity problem be addressed in a principled way?

We focus on the Probability of Direction (pd) (Makowski et al., 2019), a posterior summary that quantifies the probability that an effect falls in its dominant direction and whose use in applied psychological research has grown substantially in recent years [REF]. A key property of pd is that it can be formally linked to the one-sided frequentist p-value in standard settings (Berger & Sellke, 1987; Casella & Berger, 1987; Marsman & Wagenmakers, 2017). This correspondence makes the error consequences of thresholding pd, and therefore the multiplicity problem it generates, directly interpretable in terms familiar to most researchers.

The paper proceeds in four steps. First, we introduce pd and make its formal correspondence with the frequentist p-value explicit. Second, we characterize the null distribution of pd and examine, via simulation, how the family-wise error rate inflates under repeated thresholding. Third, we cast pd in terms of posterior odds, connect these to directional Bayes factors, and clarify the role of prior odds. Fourth, we derive a prior-odds adjustment for multiplicity control, extending the framework of Jeffreys (1938) and Westfall et al. (1997) to pd and to correlated hypotheses, and benchmark it against frequentist FWER controls (Bonferroni and Holm) (Holm, 1979). The paper concludes with a worked example and practical guidance for reporting and implementation. An accompanying R package [**REF**] implements the full procedure to facilitate its routine use in applied settings.

Probability of Direction

The probability of direction (pd) is a descriptive posterior index that quantifies how strongly the posterior distribution favors one direction of an effect relative to a reference (θ_{null}) value (Makowski et al., 2019). For a continuous posterior, it can be defined as the larger of the two one-sided posterior probabilities: $pd = \max(\Pr(\theta < \theta_{null} \mid y),\ \Pr(\theta > \theta_{null} \mid y))$, and it ranges from 0.50 (no directional preference) to 1 (all on one side). Because pd is derived from the posterior, it depends on the full model specification (likelihood and prior), so the long-run error

behavior of pd-based decisions is not a property of pd "in isolation" but of the overall Bayesian procedure that generates the posterior.

Probability of Direction as test for existence

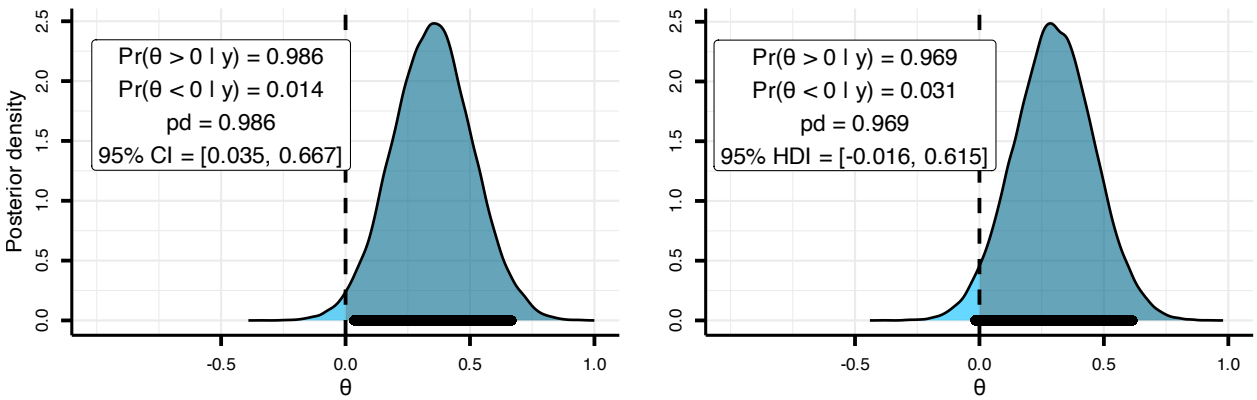
A convenient starting point for calibration is the mapping between pd and the frequentist p-value. Under flat (or sufficiently diffuse symmetric) priors in typical hypothesis-testing scenarios (e.g., testing means), pd is the posterior probability of the data-favored sign relative to θ_{null}. This value corresponds to the complement of the one-sided p-value (Casella & Berger, 1987; Marsman & Wagenmakers, 2017), yielding the two-sided approximation $p \approx 2(1 - pd)$. In practice, the pd is often used to summarize both the direction and existence of an effect [**REFs**].

This interpretation is clearest when phrased as a decision rule $pd > c$: the posterior must assign more than c probability mass to the side of θ_{null} favored by the data. Setting $c = 1 - \alpha/2$ is convenient because it bounds the posterior mass in the opposite tail by $\alpha/2$. For concreteness, suppose the posterior favors a positive effect so that $pd = \Pr(\theta > 0 \mid y)$. An $\alpha = 0.05$ implies $c = 0.975$, therefore the rule $pd > c$ requires that $\Pr(\theta \leq \theta_{\text{null}} \mid y) < 0.025$, placing less than 2.5% of posterior mass at or below 0.

Put it differently, let q_p denote the p-th posterior quantile, so the central $(1 - \alpha)$ equi-tailed credible interval is $[q_{\alpha/2},\ q_{1-\alpha/2}]$, which leaves probability $\alpha/2$ in each tail. The condition $\Pr(\theta \leq \theta_{\text{null}} \mid y) < \alpha/2$ holds if and only if θ_{null} falls below $q_{\alpha/2}$, that is, when $\theta_{\text{null}} \notin [q_{\alpha/2},\ q_{1-\alpha/2}]$. Thus, observing $pd > 0.975$ is equivalent to excluding θ_{null} from the central 95% equi-tailed credible interval (see Figure 1). When the credible interval excludes the null value, one can infer that the null is relatively implausible (Box & Tiao, 1992).

Figure 1

Posterior distributions illustrating the relationship between the probability of direction (pd) and 95% Credible Interval (CI), using $c = 0.975$ and $\theta_{null} = 0$.



Given these considerations, although pd and the frequentist p-value target different quantities, it is expected that decision rules of the form $pd > c$ approximate the operating characteristics of familiar frequentist rules such as $p < \alpha$.

Although we set $\theta_{null} = 0$ throughout the manuscript for ease of exposition, the procedure presented below readily extends to non-zero nulls. This simply requires defining directional hypotheses relative to the target value.

Multiplicity Problem

Prior impact

Under the null hypothesis, p-values are uniformly distributed on $[0, 1]$. Because pd can be mapped to the p-value, this implies a corresponding null reference behavior for pd on $[0.5, 1]$. However, because pd is a posterior probability regarding the sign of θ, its null distribution is not determined by the likelihood alone: it can shift depending on how strongly the prior concentrates probability near 0 (i.e., how strongly the prior "expects" near-null effects). Demonstrating this prior dependence at $m = 1$ makes the target of subsequent multiplicity analyses explicit. Even before any multiplicity is introduced, a fixed cutoff, such as $pd > 0.975$, does not have a universal "null meaning" across priors and sample sizes.

Given these considerations, as a first step, we simulated data under the null ($\theta = 0$) and

computed the frequency of $pd > c$, with $c = 0.975$ ($\alpha = 0.05$). Operationally, this quantifies how often the procedure falsely excludes zero from the 95% central credible interval (bounded by $1 - c$ and c), incorrectly concluding that a directed effect exists[1]. For the frequentist comparison, we conducted two-sided, one-sample z-tests (with $\alpha = 0.05$).

Specifically, we conducted a Monte Carlo simulation with 10,000 replications for sample sizes $n \in \{30, 50, 100\}$. In each replication, data were drawn from a standard normal distribution, $y_i \sim \mathcal{N}(0, 1)$. For every simulated dataset, we computed the posterior distribution of θ using a conjugate normal-normal model with $y_i \mid \theta \sim \mathcal{N}(\theta, 1)$ and prior $\theta \sim \mathcal{N}(0, \tau^2)$. We varied the prior scale $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$ to encompass a range of priors, from skeptical to relatively diffuse.

As shown in Figure 2, pd behaves analogously to a p-value when samples are large and the prior is diffuse ($\tau \geq 1$): the null distribution is approximately uniform on $[0.5, 1]$. Under skeptical priors ($\tau = 0.2$), however, the distribution concentrates away from the extremes, and the observed error rate drops.

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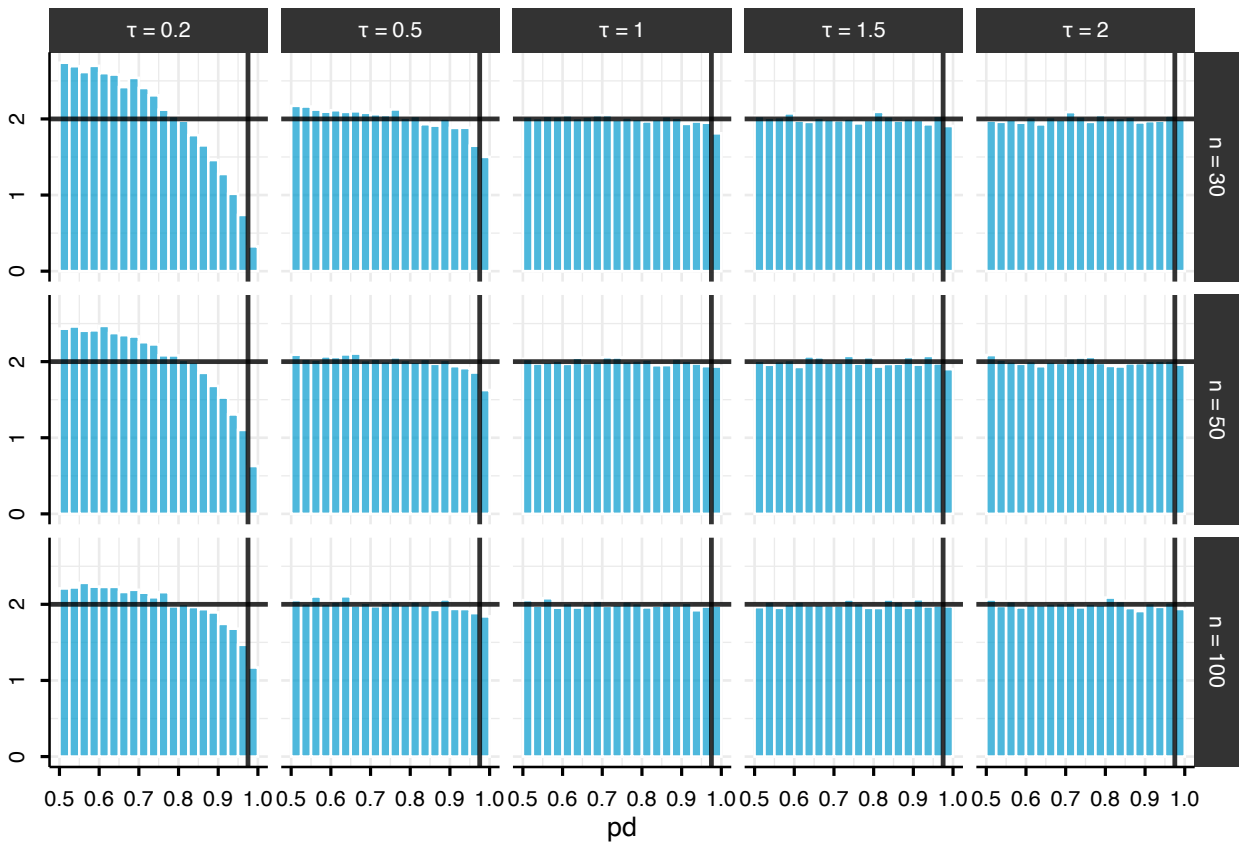
[1] In most applications, pd is estimated from posterior draws as the larger of the two one-sided posterior probabilities: the proportion of draws above θ_{null} and the proportion below θ_{null}. Because this is a Monte Carlo estimate, its numerical resolution depends directly on the number of posterior draws (Makowski et al., 2019). In our simulation, however, the conjugate normal–normal model yields a Gaussian posterior: $\theta \mid y \sim \mathcal{N}(\mu_{\text{post}}, \sigma^2_{\text{post}})$ This allows pd to be computed in closed form. Specifically, we calculate:

$pd = \max(\Pr(\theta > \theta_{\text{null}} \mid y), \Pr(\theta < \theta_{\text{null}} \mid y)) = \Phi\left(\frac{|\mu_{\text{post}} - \theta_{\text{null}}|}{\sigma_{\text{post}}}\right)$

Figure 2

Null reference distributions for the probability of direction. Empirical distributions of the probability of direction (pd) stratified by sample size ($n = 30, 50, 100$; rows) and prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$; columns). The vertical dashed line marks the commonly used evidential cutoff (pd = 0.975), which corresponds to a two-sided p-value of 0.05 under the standard conversion $p = 2(1 - pd)$.



Why skeptical priors alone are insufficient

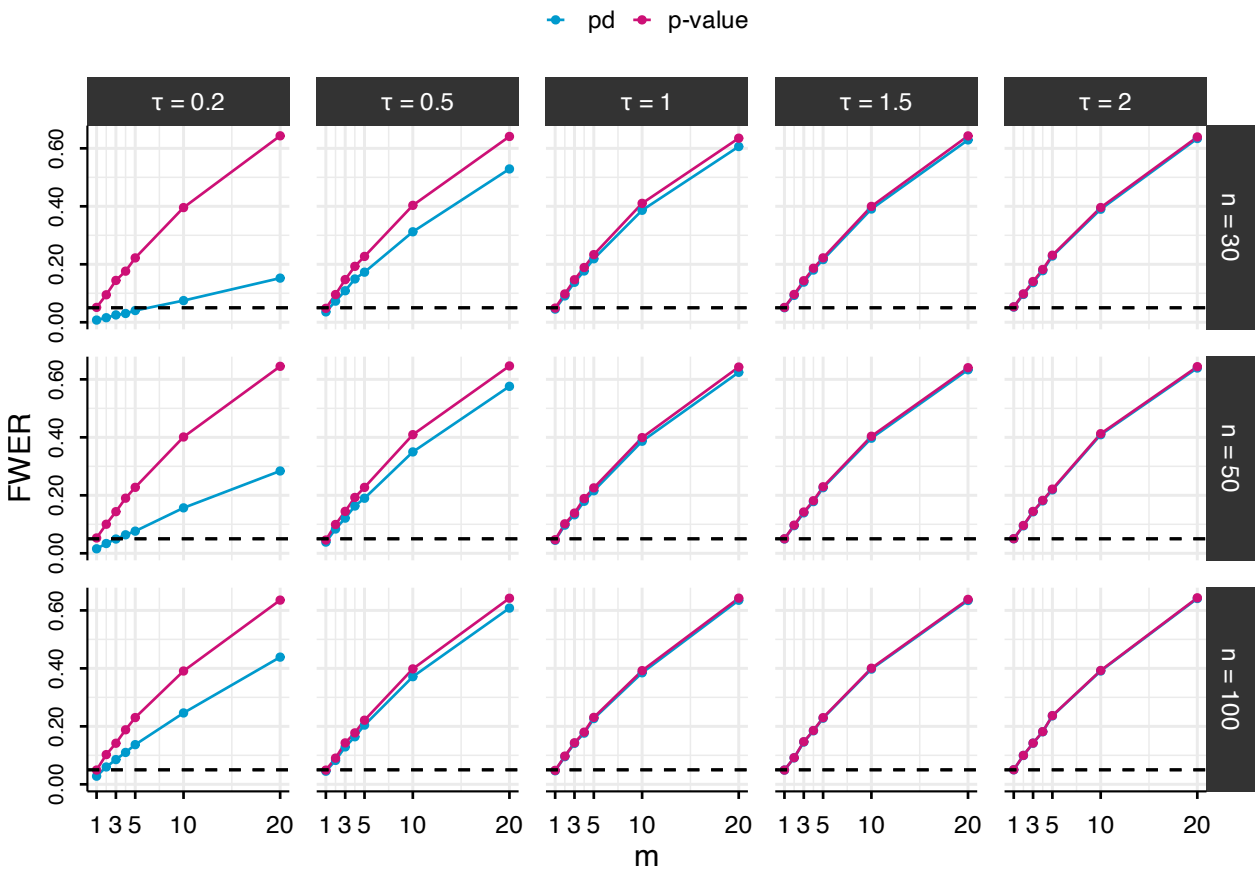
Building on this baseline, we examine the effects of multiplicity. Although individual tests rarely produce an extreme pd, the probability of at least one extreme value among $pd_1, \ldots, pd_m$ increases with the number of tests, m. We quantify this family-wise error rate (FWER) by extending the simulation to families of size $m \in \{1, 2, 3, 4, 5, 10, 20\}$ and calculating the proportion of replications with at least one $pd > 0.975$.

As expected, pd-based decisions behave broadly analogously to their frequentist

counterparts when sample sizes are large and the prior places relatively little mass near 0 (Figure 3). By contrast, as the prior becomes more skeptical (i.e., increasingly concentrated around 0), the FWER decreases. Nevertheless, the larger the sample, the less the impact of prior

Figure 3

Familywise error rates under the global null for p-values and probability of direction. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$, with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$.



The simulation results suggest that Type I error rates can be reduced by adopting more skeptical within-model priors for each parameter. However, relying on prior skepticism as a general remedy for multiplicity conflates two distinct goals: (i) encoding substantive prior knowledge about a specific parameter, and (ii) calibrating the error rate of a decision rule under repeated testing. When the primary aim is error-rate control, adjusting the within-model prior introduces bias into parameter estimates as an unintended side effect.

The approach presented below instead handles multiplicity at the level of prior odds, while leaving within-model priors unchanged. This distinguishes it from standard frequentist corrections, which operate on the decision threshold rather than on prior beliefs, and makes it

naturally compatible with the use of substantive priors. In particular, researchers can explicitly encode their expectations about how many effects are likely to be present, allowing multiplicity to be addressed without altering the underlying model. This separation allows researchers to select the most appropriate prior for each parameter based on substantive knowledge, while still controlling error rates when making joint statements across multiple parameters.

One condition, however, must be met: because pd measures the probability mass on one side of the null, the within-model prior must assign equal plausibility to effects in either direction. This symmetry requirement is satisfied by all standard default priors, and still leaves the choice of prior informativeness entirely at the researcher's discretion.

Prior-Odds adjustment for multiplicity

From posterior probabilities to posterior odds

To connect pd to posterior odds, consider the directional hypotheses $H_+ : \theta > 0$ and $H_- : \theta < 0$. These hypotheses partition the parameter space based on the sign of θ. If the prior for θ is continuous and symmetric about 0, then the two directions are equally plausible a priori: $\Pr(H_+) = \Pr(H_-) = 0.5$.

Without loss of generality, assume the posterior favors H_+, such that $pd = \Pr(H_+ \mid y) = \Pr(\theta > 0 \mid y)$. The complement is then given by $1 - pd = \Pr(\theta \leq 0 \mid y)$. The posterior odds comparing these two hypotheses are therefore:

$$\frac{pd}{1 - pd} = \frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)}$$

For a continuous posterior distribution, the probability of any single point (e.g., $\Pr(\theta = 0 \mid y)$) is 0, which means that one could write $\frac{pd}{1-pd} = \frac{\Pr(H_+|y)}{\Pr(H_-|y)}$. However, we retain the notation H_{-0} in the denominator to explicitly indicate that the hypothesis includes the point null value of zero.

Applying Bayes' rule, these posterior odds can be decomposed into a Bayes Factor and prior odds:

$$\frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)} = \mathrm{BF}_{+/-0} \times \frac{\Pr(H_+)}{\Pr(H_{-0})}$$

where $\mathrm{BF}_{+/-0}$ quantifies the relative likelihood of the data under H_+ versus H_{-0}. Under prior symmetry, $\frac{\Pr(H_+)}{\Pr(H_{-0})} = 1$, which yields $\frac{pd}{1-pd} = \mathrm{BF}_{+/-0}$ (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Thus, the odds $\frac{pd}{1-pd}$ computed from posterior draws can be interpreted directly as evidence favoring $\theta > 0$ over $\theta \leq 0$ (or $\theta < 0$ over $\theta \geq 0$).

Multiplicity adjustment via prior-odds

Assessing a single directional claim under a symmetric prior centered at θ_{null} corresponds to prior indifference (i.e., a prior probability of 0.5). If m directional claims are considered jointly and treated as independent a priori, assigning a probability of 0.5 to each claim induces a joint prior that quickly assigns high probability to the event that at least one effect is in the claimed direction.

Specifically, the prior probability that none of the m effects is in the claimed direction is $(0.5)^m$, so the prior probability that at least one is in the claimed direction is $1 - (0.5)^m$. For example, with $m = 10$, $\Pr(\text{none in claimed direction}) = 0.5^{10} \approx 0.001$, and $\Pr(\text{at least one in claimed direction}) = 1 - 0.5^{10} \approx 0.999$.

In this setting, a principled response to multiplicity is to reduce the prior odds assigned to any particular directional claim as a function of m. Because the prior is symmetric regarding the sign of θ, and pd is defined as the posterior probability of the dominant direction, the same prior odds adjustment applies whether the data favor $\theta > 0$ or $\theta < 0$.

Concretely, for each test $i = 1, \ldots, m$, let $\Pr(H_{1,i})$ denote the multiplicity-adjusted prior probability of the favored direction, and let $\Pr(H_{0,i})$ denote the prior probability of its complement (i.e., the opposite direction, with 0 included), where $\Pr(H_{0,i}) + \Pr(H_{1,i}) = 1$. Then the adjusted posterior odds satisfy:

$$\frac{pd_{\text{adj},i}}{1 - pd_{\text{adj},i}} = \frac{pd_i}{1 - pd_i} \times \frac{\Pr(H_{1,i})}{\Pr(H_{0,i})}$$

or equivalently,

$$pd_{\text{adj},i} = \frac{pd_i \Pr(H_{1,i})}{pd_i \Pr(H_{1,i}) + (1 - pd_i) \Pr(H_{0,i})}$$

Thus, multiplicity acts only through the prior odds factor $\Pr(H_{1,i})/\Pr(H_{0,i})$.

Choosing the prior odds

The multiplicity adjustment operates entirely through the prior odds. To specify these prior odds in a way that scales appropriately with the number of tests, we follow the global null formulation of Jeffreys (1938) (see also Westfall et al. (1997)). Conceptually, we first specify a prior probability for the global null: the event that all individual null hypotheses are true. We then work backward from this joint probability to derive the implied prior probability for each individual test.

Formally, for each test $i = 1, \ldots, m$, let $H_{1,i}$ be the directional claim corresponding to the posterior-dominant sign (i.e., the direction used to compute $pd_i = \Pr(H_{1,i} \mid y)$), and let $H_{0,i}$ denote its complement (which includes zero). Note that $H_{1,i}$ is defined post hoc based on the observed posterior direction.

The global null hypothesis, $\mathbf{H_0}$, is the event that none of the m parameters lie in their respective favored directions:

$$\mathbf{H_0} = \bigcap_{i=1}^{m} H_{0,i}$$

Assuming prior independence across hypotheses and a common per-hypothesis prior probability $\Pr(H_{1,i}) = k$, the probability of the global null is given by $\Pr(\mathbf{H_0}) = (1 - k)^m$. Solving for k yields the adjusted prior probability for each individual test:

$$k = 1 - \Pr(\mathbf{H_0})^{1/m}$$

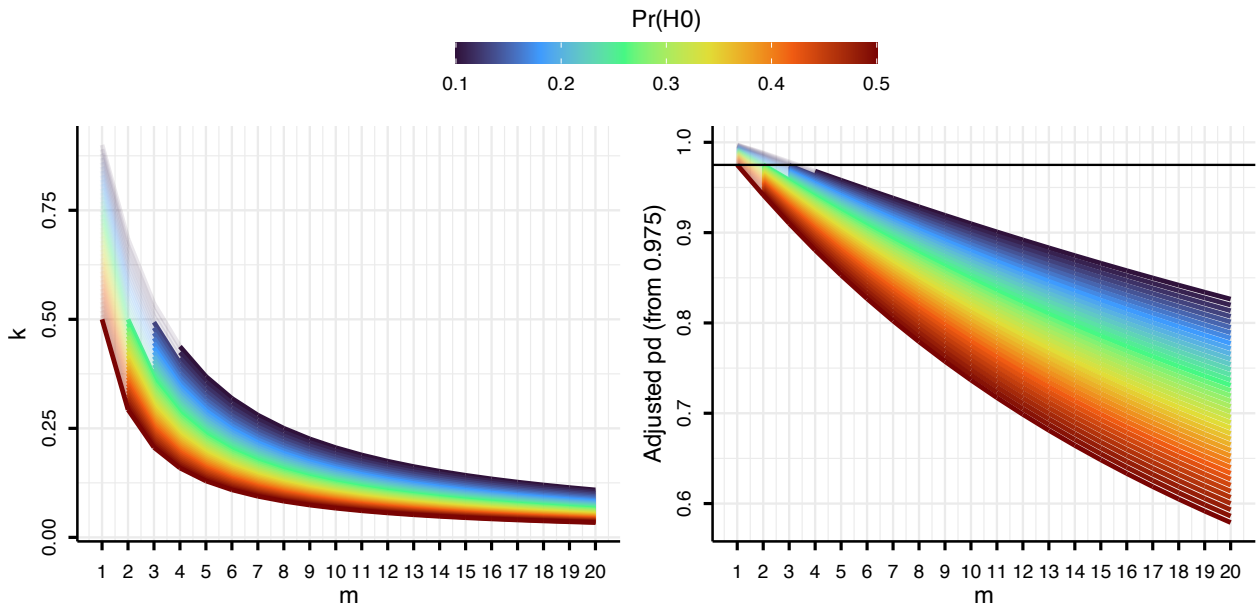
This formulation ensures the multiplicity penalty scales logically. For example, if a researcher tests $m = 5$ hypotheses and assigns a prior probability of $\Pr(\mathbf{H_0}) = 0.5$ to the global null, the prior probability for any single test finding an effect in the expected direction becomes $k = 1 - 0.5^{1/5} \approx 0.129$. As the number of tests (m) increases, k decreases, rendering the prior increasingly skeptical of any single claim.

Researchers can encode their knowledge by selecting $\Pr(\mathbf{H_0})$. When few non-null effects are anticipated, $\Pr(\mathbf{H_0})$ should be set relatively high (e.g., 0.4); conversely, when effects are well established, $\Pr(\mathbf{H_0})$ can be lower (e.g., 0.1). Once $\Pr(\mathbf{H_0})$ is fixed, larger families necessarily

imply a smaller k, thereby placing a higher prior probability on the null for each individual test (Figure 4).

Figure 4

Impact of the multiplicity penalty. Left: The implied prior probability $k = \Pr(H_{1,i})$ *assigned to each hypothesis* $H_{1,i}$ *decreases as the family size* m *grows, depending on the specified global null probability* $\Pr(\mathbf{H_0})$*. The faded region represents boundary conditions where the calculated* k *exceeds* 0.5*. Right: How this prior calibration translates to the probability of direction. Starting from unadjusted probability of direction (*$pd = 0.975$*, black line), the adjusted* pd *is penalized more heavily as both* m *and* $\Pr(\mathbf{H_0})$ *increase.*



However, a boundary condition arises when the family of tests is small (m) and researchers hold a strong a priori expectation of non-null effects (i.e., a very low $\Pr(\mathbf{H_0})$). Under these conditions, the calibration can become anti-conservative: the formula may assign a prior probability $k > 0.5$ to the favored direction (Figure 4, left panel). Because this would artificially inflate the posterior odds rather than penalizing them (Figure 4, right panel), we impose a strict upper bound. If the formula yields $k > 0.5$, we cap it at 0.5 (ensuring $\Pr(H_{0,i}) \geq 0.5$). This guarantees the adjustment acts exclusively as a conservative safeguard against false positives, leaving the posterior probabilities unchanged from their unadjusted baseline when no penalty is

warranted.

Simulation Study 1

To illustrate the impact of $\Pr(\mathbf{H_0})$ on both FWER and power, we simulated two data-generating regimes: a global-null scenario ($\theta_1 = \cdots = \theta_m = 0$) and an alternative scenario where exactly one effect is non-null ($\theta_1 = \cdots = \theta_{m-1} = 0, \theta_m = 0.3$). Replicating our previous simulation procedure with a fixed sample size of $n = 50$, we varied the global null prior across $\Pr(\mathbf{H_0}) \in \{0.50, 0.40, 0.30, 0.20, 0.10\}$. Following Westfall et al. (1997), we compared this Bayesian multiplicity adjustment with the Bonferroni p-value correction, which provides strong family-wise error rate (FWER) control.

Figure 5

Family-wise error rate (FWER) control as a function of family size (m) for the Bayesian adjusted pd (blue) and frequentist Bonferroni (pink) correction methods. Results are paneled by the global null prior probability, $\Pr(\mathbf{H_0})$*, and the prior scale,* τ*, for simulated datasets of size* $n = 50$*. The horizontal dashed line indicates the nominal* $\alpha = 0.05$ *threshold.*

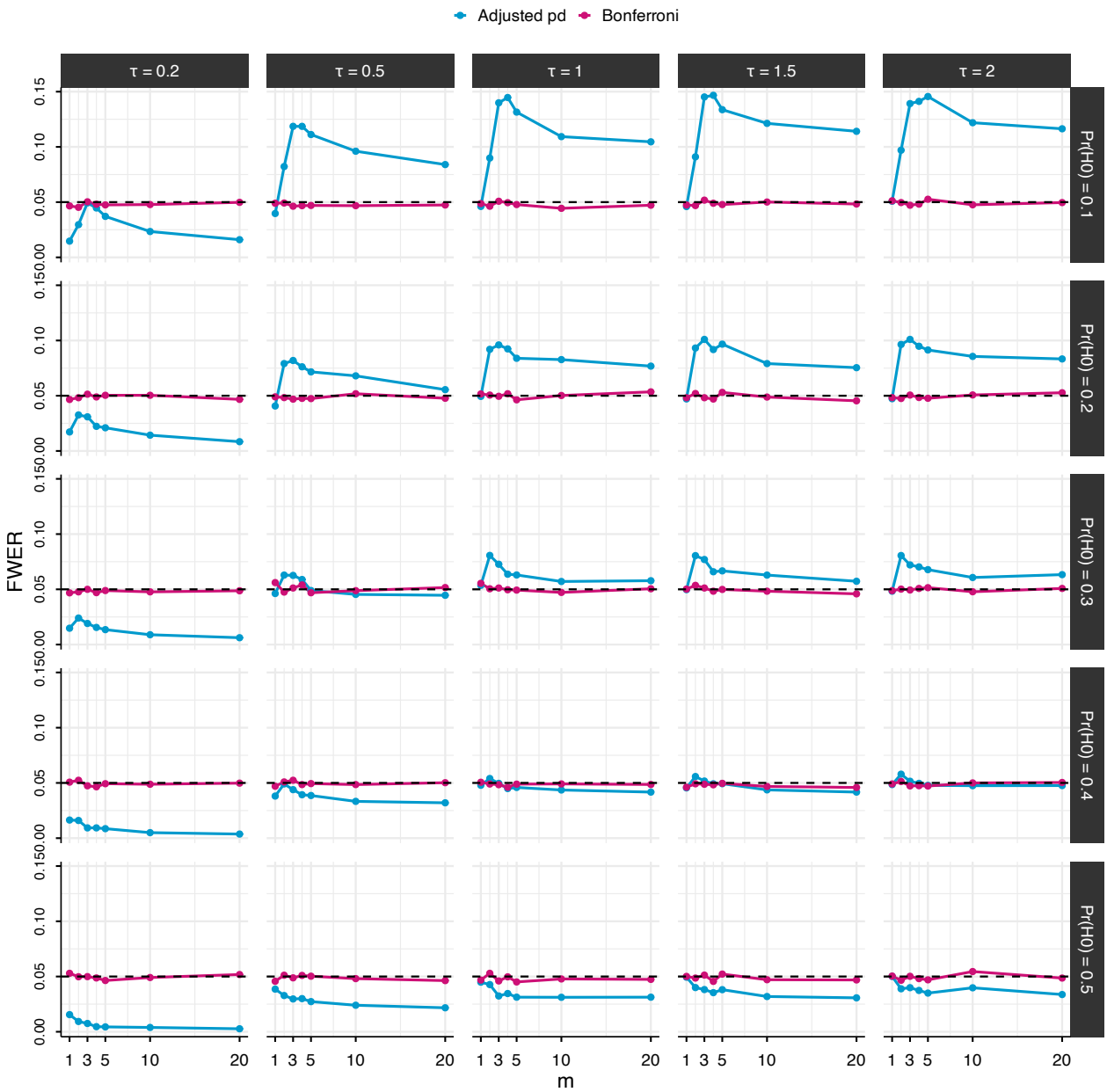
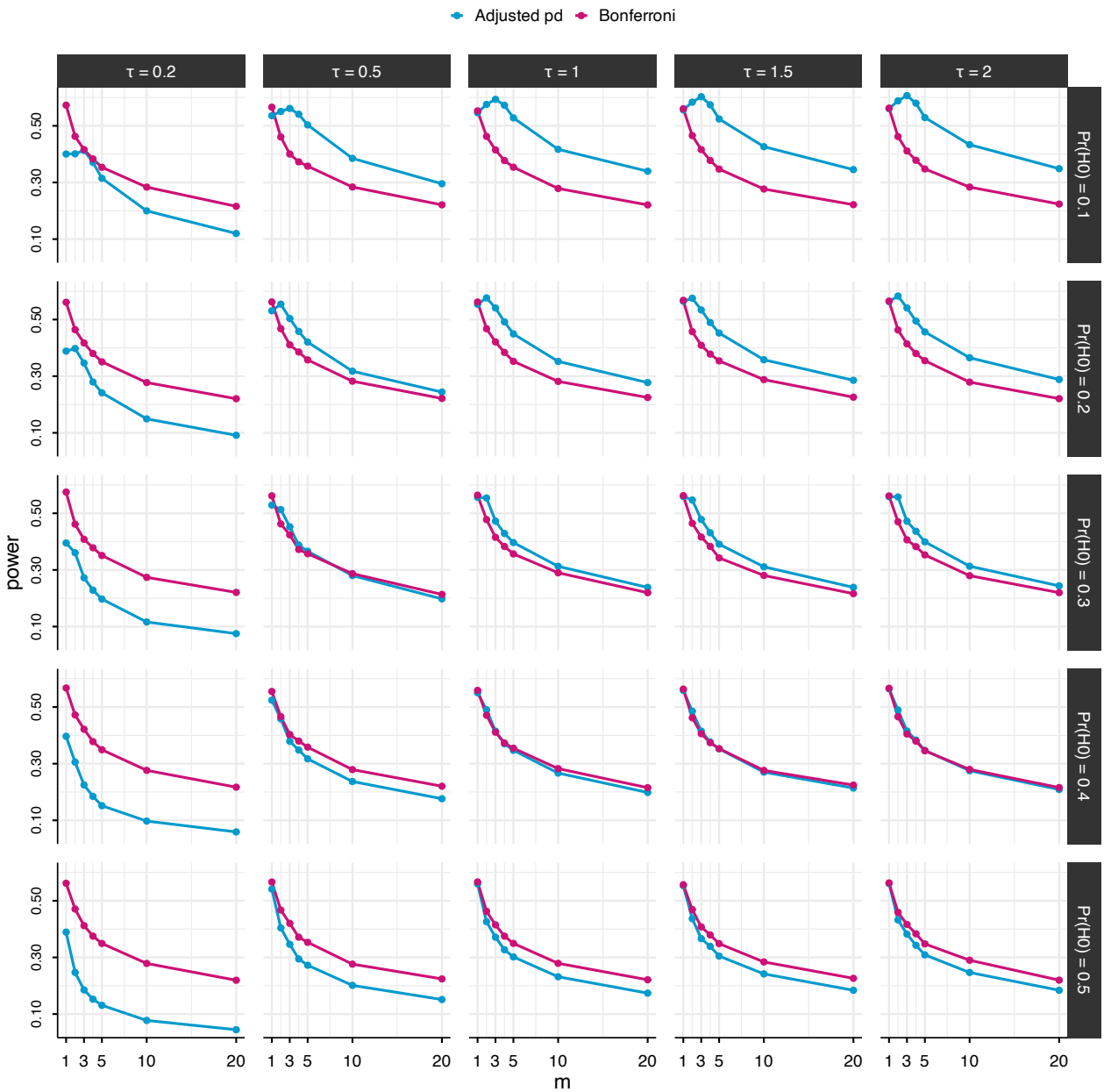


Figure 6

Power as a function of family size (m) for the Bayesian adjusted pd (blue) and frequentist Bonferroni (pink) correction methods. Results are shown across varying prior scales (τ) and global null prior probabilities, $\Pr(\mathbf{H_0})$*, for* $n = 50$.



Of foremost importance, if highly skeptical priors are already employed for estimation (e.g., $\tau = 0.2$), applying this multiplicity adjustment constitutes a restrictive "double" correction. Beyond this case, the adjusted *pd* procedure offers a flexible trade-off governed by the global null

prior parameter, $\Pr(\mathbf{H_0})$.

An area of equivalence emerges when setting $\Pr(\mathbf{H_0}) = 0.4$, which produces FWER and power trajectories that closely mirror the Bonferroni adjustment across varying family sizes. This empirical alignment is consistent with the asymptotic proof by Westfall et al. (1997) demonstrating that a global null probability of $\Pr(\mathbf{H_0}) = e^{-1} \approx 0.368$ reproduces Bonferroni-like behavior. Thus, selecting $\Pr(\mathbf{H_0}) \approx 0.4$ effectively calibrates the Bayesian prior to match the severity of the frequentist α correction.

The simulation results further reveal that when researchers hold a strong expectation of non-null effects (e.g., $\Pr(\mathbf{H_0}) \leq 0.2$), the Bayesian adjustment yields higher power than the Bonferroni correction but suffers from FWER inflation, particularly as the prior scale (τ) increase (Figure 5). Conversely, imposing a highly skeptical global prior ($\Pr(\mathbf{H_0}) = 0.5$) over-corrects the FWER to well below the nominal 0.05 threshold, correspondingly sacrificing power (Figure 6).

Lastly, although setting $\Pr(\mathbf{H_0}) \in \{0.2, 0.3\}$ yields FWER and power trajectories distinct from the Bonferroni method, these priors still provide substantial error control. Specifically, they maintain a maximum FWER of roughly 10% (only 5% above the nominal level, with $m = 3$ and $\Pr(\mathbf{H_0}) = 0.2$) while delivering a notable power increase of approximately +15%.

Accounting for correlation

A limitation of the proposed procedure is that it allocates the global-null probability across m hypotheses under an independence assumption. When hypotheses are correlated, for instance because parameters are estimated jointly within a single model or because measures share the same observations, independence-based multiplicity adjustments tend to be overly conservative (Pesarin & Salmaso, 2010; Westfall & Young, 1993).

One principled remedy is to replace the nominal number of tests m with an effective number m_{eff} that reflects the amount of unique information in the family. This requires estimating a correlation matrix $\mathbf{R}$ among the tests (e.g., among posterior distributions) and mapping it to m_{eff} via an eigenvalue decomposition. Several methods for this mapping have been proposed, each resting on different assumptions (Cheverud, 2001; Li & Ji, 2005; see Del Giudice, 2021 for a

critical discussion); here we adopt the eigenvalue-based formula of Cheverud (2001). This correction can be applied post hoc given only an estimated or assumed correlation matrix[2].

Simulation Study 2

To evaluate this approach, we extended the simulation study to correlated settings. We generated families of $m \in \{2, 4, 6, 10, 20\}$ hypotheses with $n = 50$ observations per dataset, across 10,000 replications per condition. In each replication, data were generated from a multivariate normal distribution with an equicorrelation structure: $y_i \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with $\mu_1 = \cdots = \mu_{m/2} = 0$ (null hypotheses) and $\mu_{m/2+1} = \cdots = \mu_m = 0.3$ (non-null hypotheses), corresponding to a scenario in which exactly half the hypotheses are true nulls. The off-diagonal entries of $\mathbf{R}$ were set to a constant $\rho \in \{0.2, 0.4, 0.6, 0.8\}$.

Posterior distributions were obtained using a multivariate normal conjugate model with prior mean $\mu = 0$ and prior standard deviation $\tau = 2$, a specification under which Bayesian and frequentist inference coincide. The prior weight on the global null was fixed at $\Pr(\mathbf{H_0}) = 0.4$, consistent with earlier results. Performance was evaluated in terms of FWER within the null subset and average power, defined as the proportion of correctly identified non-null hypotheses (Benjamini & Hochberg, 1995; Izmirlian et al., 2025).

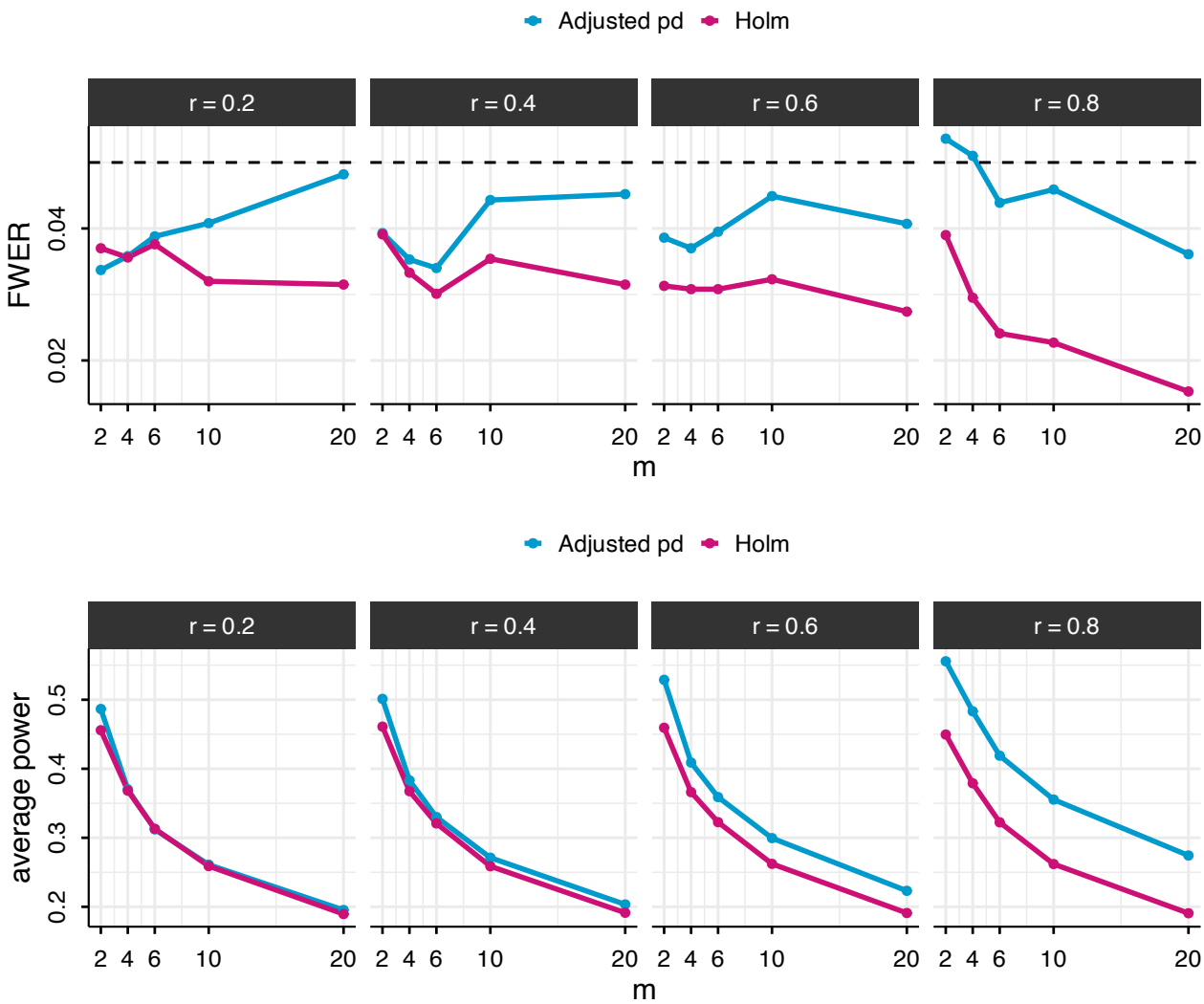
We compared the pd-based multiplicity correction against the Holm procedure as a frequentist family-wise error rate benchmark (Holm, 1979). Holm was selected over Bonferroni because it provides strong FWER control while achieving greater power, and it is among the most widely implemented correction procedures in applied research.

[2] We also provide the option of supplying a single scalar representing the expected average correlation among tests.

As shown in Figure 7, independence-based corrections are overconservative when tests are correlated, as is well known, leading to a substantial loss of power. By contrast, the m_{eff}-based procedure maintained nominal FWER control while retaining greater power, with the advantage growing as ρ increased.

Figure 7

Familywise error rate (top panels) and average power (bottom panels) for the adjusted probability of direction methods as a function of the number of tests (m) and correlation structure (r). The adjusted probabilit incorporates an effective number of tests correction to account for correlation structure. Each column represents a different correlation level.



Application: worked example with the R Package (in progress)

The procedure is implemented in the `multibayes` R package, which provides functions for computing adjusted pd values from either raw posterior draws or a vector of unadjusted pd values. The function developed from this work, `pd.adjust()`, ….

```
# install.packages("remotes")
remotes::install_github("mar-cald/multibayes")
library(multibayes)
```

The output returns the adjusted pd values alongside the implied global null probability and the effective number of tests m_{eff} (when a correlation matrix is supplied). The unadjusted values are also retained for direct comparison.

Discussion

Although pd is a descriptive posterior index, thresholding it into a decision rule and applying that rule repeatedly across many parameters induces a multiplicity problem. Without adjustment, researchers implicitly accept a prior probability for the global null hypothesis that may be far lower than intended. This concern is particularly acute when non-informative or weakly informative priors are used, or when sufficient data render the likelihood dominant. The procedure proposed here targets this decision-level problem by extending the foundational work of Jeffreys (1938) and Westfall et al. (1997) to the pd context, while allowing researcher-specific knowledge to enter transparently through the prior-odds specification. Researchers can explicitly encode their expectations about the prevalence of true effects through the prior probability assigned to the global null hypothesis, separating within-model priors on parameters from prior assumptions over the hypothesis space. In this sense, the procedure is not merely a correction for multiple testing, but a transparent mechanism for connecting prior beliefs to decision-making under multiplicity.

Related corrections have been implemented for specific testing contexts, such as post-hoc tests in JASP (Jong, 2019; van den Bergh et al., 2020). The present procedure extends this logic to pd, accounts for correlations among tested hypotheses, and provides a general-purpose implementation for applied use. Unlike existing implementations tied to specific software environments or model types, it applies to pd values arising from any modeling strategy. While the most coherent approach analyzes all relevant parameters within a single joint model that explicitly represents dependence (Gelman et al., 2012, 2013; Kruschke, 2011), this is not always feasible in practice and researchers often report pd values from multiple separate analyses within the same manuscript [REFs]. The procedure accommodates all settings: when a joint model is available, its posterior correlation structure can be leveraged directly; when only separate analyses are available, the adjustment proceeds under an independence assumption; when partial information is available, one can supply assumed correlations for specific parameter pairs or a single scalar representing the expected mean correlation across tests.

An important caveat is that this framework does not provide evidence for a point null

hypothesis (i.e., posterior support for θ_{null}). Instead, it evaluates directional claims based on inequalities (e.g., $\theta > \theta_{null}$ vs. $\theta \leq \theta_{null}$). By construction, the complementary hypothesis includes the null value, but the framework does not assign it special status. When the null value falls outside the credible interval, one can infer its relative implausibility as a candidate value, but this differs fundamentally from quantifying evidence in its favor. Researchers whose primary interest lies in null evidence are better served by Bayes factor approaches that assign the null positive prior mass [Heck et al. (2023); Morey and Rouder (2011); Ly et al. (2016); add dienes].

Several further limitations should be noted. The formal correspondence between pd and the frequentist p-value holds under specific conditions, that is, continuous posteriors, sufficiently large samples, and/or diffuse priors, and may not generalize to small-sample settings or analyses with strongly informative priors. The m_{eff} correction additionally requires estimating or assuming the correlation structure among tests; misspecification of this structure will propagate into the adjustment. Finally, the framework is developed specifically for pd, but extending the prior-odds logic to other posterior indices is a natural direction for future work.

For applied reporting, we recommend stating the number of tests constituting the family and the implied prior probability under the chosen correction. The unadjusted pd should be reported for each parameter alongside at least one multiplicity-adjusted measure based on the proposed prior-odds correction.

As Bayesian methods continue to be adopted in psychological research, often in pragmatic settings where default priors and conventional thresholds are the norm, tools that make the error consequences of such practices explicit, and offer principled corrections, become increasingly necessary. The procedure developed here provides a coherent and practical way to recalibrate pd-based decisions for multiple tests, making reported claims less likely to reflect chance patterns across many parameters.

References

Albers, C. J., Kiers, H. A., & Ravenzwaaij, D. van. (2018). Credible confidence: A pragmatic view on the frequentist vs Bayesian debate. *Collabra: Psychology*, *4*(1).

Altoè, G., Bertoldo, G., Zandonella Callegher, C., Toffalini, E., Calcagnì, A., Finos, L., & Pastore, M. (2020). Enhancing statistical inference in psychological research via prospective and retrospective design analysis. *Frontiers in Psychology*, *Volume 10 - 2019*. https://doi.org/10.3389/fpsyg.2019.02893

Bender, R., & Lange, S. (2001). Adjusting for multiple testing—when and how? *Journal of Clinical Epidemiology*, *54*(4), 343–349. https://doi.org/https://doi.org/10.1016/S0895-4356(00)00314-0

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, *57*(1), 289–300. https://doi.org/https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Berger, J. O., & Sellke, T. (1987). Testing a point null hypothesis: The irreconcilability of p values and evidence. *Journal of the American Statistical Association*, *82*(397), 112–122.

Bernardo, J. M., & Smith, A. F. M. (1994). *Bayesian Theory* (1st ed.). Wiley. https://doi.org/10.1002/9780470316870

Berry, D. A., & Hochberg, Y. (1999). Bayesian perspectives on multiple comparisons. *Journal of Statistical Planning and Inference*, *82*(1), 215–227. https://doi.org/10.1016/S0378-3758(99)00044-0

Box, G. E. P. (1980). Sampling and bayes' inference in scientific modelling and robustness. *Journal of the Royal Statistical Society. Series A (General)*, *143*(4), 383–430. http://www.jstor.org/stable/2982063

Box, G. E. P., & Tiao, G. C. (1992). *Bayesian inference in statistical analysis* (1st ed.). Wiley. https://doi.org/10.1002/9781118033197

Casella, G., & Berger, R. L. (1987). Reconciling bayesian and frequentist evidence in the

one-sided testing problem. *Journal of the American Statistical Association*, *82*(397), 106–111. https://doi.org/10.1080/01621459.1987.10478396

Chen, G., Taylor, P. A., Cox, R. W., & Pessoa, L. (2020). Fighting or embracing multiplicity in neuroimaging? Neighborhood leverage versus global calibration. *NeuroImage*, *206*, 116320. https://doi.org/10.1016/j.neuroimage.2019.116320

Cheverud, J. M. (2001). A simple correction for multiple comparisons in interval mapping genome scans. *Heredity*, *87*(1), 52–58. https://doi.org/https://doi.org/10.1046/j.1365-2540.2001.00901.x

Cohen, J. (1994). The earth is round (p < .05). *American Psychologist*, *49*(12), 997–1003. https://doi.org/10.1037/0003-066X.49.12.997

Cramer, A. O. J., van Ravenzwaaij, D., Matzke, D., Steingroever, H., Wetzels, R., Grasman, R. P. P. P., Waldorp, L. J., & Wagenmakers, E.-J. (2016). Hidden multiplicity in exploratory multiway ANOVA: Prevalence and remedies. *Psychonomic Bulletin & Review*, *23*(2), 640–647. https://doi.org/10.3758/s13423-015-0913-5

Dawid, A. P. (1994). Selection paradoxes of bayesian inference. *Lecture Notes-Monograph Series*, *24*, 211–220.

Del Giudice, M. (2021). Effective dimensionality: A tutorial. *Multivariate Behavioral Research*, *56*(3), 527–542.

Edwards, W., Lindman, H., & Savage, L. J. (1963). Bayesian statistical inference for psychological research. *Psychological Review*, *70*(3), 193.

Erčulj, F., & Štrumbelj, E. (2015). Basketball shot types and shot success in different levels of competitive basketball. *PLOS ONE*, *10*(6), 1–14. https://doi.org/10.1371/journal.pone.0128885

Gelman, A., & Carlin, J. (2014). Beyond power calculations: Assessing type s (sign) and type m (magnitude) errors. *Perspectives on Psychological Science*, *9*(6), 641–651. https://doi.org/10.1177/1745691614551642

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013).

Bayesian data analysis (3rd ed.). Chapman; Hall/CRC. https://doi.org/10.1201/b16018

Gelman, A., & Hennig, C. (2017). Beyond subjective and objective in statistics. *Journal of the Royal Statistical Society Series A: Statistics in Society*, *180*(4), 967–1033. https://doi.org/10.1111/rssa.12276

Gelman, A., Hill, J., & Yajima, M. (2012). Why we (usually) don't have to worry about multiple comparisons. *Journal of Research on Educational Effectiveness*, *5*(2), 189–211. https://doi.org/10.1080/19345747.2011.618213

Gelman, A., & Shalizi, C. R. (2013). Philosophy and the practice of bayesian statistics. *British Journal of Mathematical and Statistical Psychology*, *66*(1), 8–38. https://doi.org/https://doi.org/10.1111/j.2044-8317.2011.02037.x

Gelman, A., & Tuerlinckx, F. (2000). Type s error rates for classical and bayesian single and multiple comparison procedures. *Computational Statistics*, *15*(3), 373–390. https://doi.org/10.1007/s001800000040

Gigerenzer, G. (2004). Mindless statistics. *The Journal of Socio-Economics*, *33*(5), 587–606. https://doi.org/https://doi.org/10.1016/j.socec.2004.09.033

Heck, D. W., Boehm, U., Böing-Messing, F., Bürkner, P.-C., Derks, K., Dienes, Z., Fu, Q., Gu, X., Karimova, D., Kiers, H. A. L., Klugkist, I., Kuiper, R. M., Lee, M. D., Leenders, R., Leplaa, H. J., Linde, M., Ly, A., Meijerink-Bosman, M., Moerbeek, M., … Hoijtink, H. (2023). A review of applications of the Bayes factor in psychological research. *Psychological Methods*, *28*(3), 558–579. https://doi.org/10.1037/met0000454

Hogeveen, J., Mullins, T. S., Romero, J. D., Eversole, E., Rogge-Obando, K., Mayer, A. R., & Costa, V. D. (2022). The neurocomputational bases of explore-exploit decision-making. *Neuron*, *110*(11), 1869–1879.e5. https://doi.org/10.1016/j.neuron.2022.03.014

Hoijtink, H., Kooten, P. van, & Hulsker, K. (2016). Why bayesian psychologists should change the way they use the bayes factor. *Multivariate Behavioral Research*, *51*(1), 2–10.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, *6*(2), 65–70. http://www.jstor.org/stable/4615733

Izmirlian, G., Sirota, L. A., Berger, V. W., & Kipnis, V. (2025). The fundamentals of multiplicity adjustment in biostatistics. *JNCI Monographs*, *2025*(68), 10–13. https://doi.org/10.1093/jncimonographs/lgae050

Jeffreys, H. (1938). Significance tests when several degrees of freedom arise simultaneously. *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, *165*(921), 161–198. https://doi.org/10.1098/rspa.1938.0052

Jevremov, T., & Pajić, D. (2024). Bayesian method in psychology: A bibliometric analysis. *Current Psychology*, *43*(10), 8644–8654.

Jong, T. de. (2019). *A bayesian approach to the correction for multiplicity*. PsyArXiv preprint. https://doi.org/10.31234/osf.io/s56mk

Kelter, R. (2023). How to choose between different bayesian posterior indices for hypothesis testing in practice. *Multivariate Behavioral Research*, *58*(1), 160–188.

Kruschke, J. K. (2011). Bayesian assessment of null values via parameter estimation and model comparison. *Perspectives on Psychological Science*, *6*(3), 299–312. https://doi.org/10.1177/1745691611406925

Kruschke, J. K. (2018). Rejecting or Accepting Parameter Values in Bayesian Estimation. *Advances in Methods and Practices in Psychological Science*, *1*(2), 270–280. https://doi.org/10.1177/2515245918771304

Li, J., & Ji, L. (2005). Adjusting multiple testing in multilocus analyses using the eigenvalues of a correlation matrix. *Heredity*, *95*(3), 221–227.

Limbachia, C., Morrow, K., Khibovska, A., Meyer, C., Padmala, S., & Pessoa, L. (2021). Controllability over stressor decreases responses in key threat-related brain areas. *Communications Biology*, *4*(1), 42. https://doi.org/10.1038/s42003-020-01537-5

Lindley, D. V. (1972). Bayesian statistics a review. In *Bayesian statistics* (pp. 1–74). Society for Industrial; Applied Mathematics. https://doi.org/10.1137/1.9781611970654.ch1

Ly, A., Verhagen, J., & Wagenmakers, E.-J. (2016). Harold jeffreys's default bayes factor hypothesis tests: Explanation, extension, and application in psychology. *Journal of*

Mathematical Psychology, *72*, 19–32. https://doi.org/https://doi.org/10.1016/j.jmp.2015.06.004

Makowski, D., Ben-Shachar, M. S., Chen, S. H. A., & Lüdecke, D. (2019). Indices of effect existence and significance in the bayesian framework. *Frontiers in Psychology*, *Volume 10 - 2019*. https://doi.org/10.3389/fpsyg.2019.02767

Marsman, M., & Wagenmakers, E.-J. (2017). Three insights from a bayesian interpretation of the one-sided p value. *Educational and Psychological Measurement*, *77*(3), 529–539. https://doi.org/10.1177/0013164416669201

Mathar, D., Erfanian Abdoust, M., Marrenbach, T., Tuzsus, D., & Peters, J. (2022). The catecholamine precursor tyrosine reduces autonomic arousal and decreases decision thresholds in reinforcement learning and temporal discounting. *PLOS Computational Biology*, *18*(12). https://doi.org/10.1371/journal.pcbi.1010785

Mayo, D. G. (2018). *Statistical inference as severe testing: How to get beyond the statistics wars*. Cambridge University Press.

Morey, R. D., & Rouder, J. N. (2011). Bayes factor approaches for testing interval null hypotheses. *Psychological Methods*, *16*(4), 406–419. https://doi.org/10.1037/a0024377

Pesarin, F., & Salmaso, L. (2010). The permutation testing approach: A review. *Statistica*, *70*(4), 481–509. https://doi.org/10.6092/issn.1973-2201/3599

Pfadt, J. M., Bartoš, F., Godmann, H. R., Waaijers, M., Groot, L., Heo, I., et al. (2025, October 30). *A methodological metamorphosis: The rapid rise of bayesian inference and open science practices in psychology*. OSF Preprints. https://doi.org/10.31234/osf.io/ck3js_v1

Pruzek, R. M. (2016). *An introduction to bayesian inference and its applications* (pp. 305–334). Routledge.

Rozeboom, W. W. (1960). The fallacy of the null-hypothesis significance test. *Psychological Bulletin*, *57*(5), 416–428. https://doi.org/10.1037/h0042040

Schnell, P. M., Fiecas, M., & Carlin, B. P. (2020). **credsubs** : Multiplicity-adjusted subset identification. *Journal of Statistical Software*, *94*(7). https://doi.org/10.18637/jss.v094.i07

Scott, J. G., & Berger, J. O. (2006). An exploration of aspects of bayesian multiple testing. *Journal of Statistical Planning and Inference*, *136*(7), 2144–2162.

Scott, J. G., & Berger, J. O. (2010). Bayes and empirical-Bayes multiplicity adjustment in the variable-selection problem. *The Annals of Statistics*, *38*(5), 2587–2619. https://doi.org/10.1214/10-AOS792

Smid, S. C., & Winter, S. D. (2020). Dangers of the defaults: A tutorial on the impact of default priors when using bayesian SEM with small samples. *Frontiers in Psychology*, *Volume 11 - 2020*. https://doi.org/10.3389/fpsyg.2020.611963

Tendeiro, J. N., & Kiers, H. A. (2019). A review of issues about null hypothesis bayesian testing. *Psychological Methods*, *24*(6), 774.

Tendeiro, J. N., Kiers, H. A., Hoekstra, R., Wong, T. K., & Morey, R. D. (2024). Diagnosing the misuse of the bayes factor in applied research. *Advances in Methods and Practices in Psychological Science*, *7*(1), 25152459231213371.

Van De Schoot, R., Winter, S. D., Ryan, O., Zondervan-Zwijnenburg, M., & Depaoli, S. (2017). A systematic review of bayesian articles in psychology: The last 25 years. *Psychological Methods*, *22*(2), 217.

van den Bergh, D., van Doorn, J., Marsman, M., Draws, T., van Kesteren, E.-J., Derks, K., Dablander, F., Gronau, Q. F., Kucharský, Š., Gupta, A. R. K. N., Sarafoglou, A., Voelkel, J. G., Stefan, A., Ly, A., Hinne, M., Matzke, D., & Wagenmakers, E.-J. (2020). A tutorial on conducting and interpreting a Bayesian ANOVA in JASP. *L'Année Psychologique*, *120*(1), 73–96. https://doi.org/10.3917/anpsy1.201.0073

Westfall, P. H., Johnson, W. O., & Utts, J. M. (1997). A bayesian perspective on the bonferroni adjustment. *Biometrika*, *84*(2), 419–427.

Westfall, P. H., & Young, S. S. (1993). *Resampling-based multiple testing: Examples and methods for p-value adjustment*. Wiley.

Zeevi, Y., Catzman, S., Benjamini, Y., & Mudrik, L. (2025). Correction for multiple comparisons should be ubiquitous. *Nature Human Behaviour*, *9*(12), 2407–2408.

https://doi.org/10.1038/s41562-025-02313-3

Zwet, E. van, & Cator, E. A. (2021). The significance filter, the winner's curse and the need to shrink. *Statistica Neerlandica*, *75*(4), 437–452. https://doi.org/10.1111/stan.12241

Zwet, E. van, & Gelman, A. (2022). A proposal for informative default priors scaled by the standard error of estimates. *The American Statistician*, *76*(1), 1–9. https://doi.org/10.1080/00031305.2021.1938225